Homework 04
REAL VARIABLES I
Fall 2026

Due: 2026-09-20 19:00
Topics: Completeness, product measures, and metric outer measures

Instructions
Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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1. (Completeness)

1.a (weight: 0.3) Let $(X, \mathcal{E}, \mu)$ be a measure space. Prove that the following two definitions of completeness are equivalent:

1. Every subset of every set $N \in \mathcal{E}$ with $\mu(N) = 0$ belongs to $\mathcal{E}$.
2. Whenever $A \in \mathcal{E}$, $B \subseteq X$, and
$$A \triangle B \subseteq N \qquad \text{for some } N \in \mathcal{E} \text{ with } \mu(N) = 0,$$
then $B \in \mathcal{E}$. Recall that $A \Delta B \coloneqq (A \cup B) \setminus (A \cap B)$.

1.b (weight: 0.7) Let μ^* be an outer measure on X, let $\mathcal{M}_{\mu^*}$ be its Carathéodory measurable sets, and let $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$. Prove that the measure space $(X, \mathcal{M}_{\mu^*}, \mu)$ is complete.

2. (Distribution functions and probability measures)

Let $\mathcal{D}$ consist of the nondecreasing functions $F\colon [0,1] \to [0,1]$ satisfying $F(0) = 0$, $F(1) = 1$, and
$$F(t) = \lim_{s \uparrow t} F(s) \qquad \text{for every } 0 < t \leq 1.$$

Let $\mathcal{P}([0,1))$ be the probability Borel measures on $[0,1)$. We will construct the bijection
$$\mathcal{D} \longleftrightarrow \mathcal{P}([0,1)), \qquad F \longmapsto \mu_F, \qquad \mu \longmapsto F_\mu,$$

where

$$F_\mu(t) \coloneqq \mu([0,t)) \quad (0 \le t < 1), \qquad F_\mu(1) \coloneqq 1.$$

2.a (weight: 0.4)
Prove that given μ the function F_μ satisfies the desired properties. Prove that F_μ is discontinuous at $t \in [0,1)$ if and only if $\mu(\{t\}) > 0$.

2.b (weight: 0.1) Prove uniqueness: if $F_\mu = F_\nu$ then $\mu = \nu$.

2.c (weight: 0.5) IN CLASS
Use the following outline to prove the claim.
Let $\mathcal{I} \coloneqq \{[a,b) : 0 \le a \le b \le 1\}$. Find the correct way to define the cost $l([a,b))$ based on F. Next, let l^* be the outer measure generated by l.
Prove that it extends l. Be very careful about how you pass from a countable to a finite cover.
Next, invoke Caratheodory's theorem to obtain a σ algebra on which l^* is a measure.
Prove $\mathcal{I}$ are measurable.
Summarize the construction and show how you proved the claim

3. (Borel sets for a metric outer measure)

3.a (weight: 1) IN CLASS
Let (X,d) be a metric space, and let μ^* be an outer measure on X with the following property:

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \qquad \text{whenever } \operatorname{dist}(A,B) > 0,$$

where

$$\operatorname{dist}(A,B) \coloneqq \inf\{d(a,b) : a \in A,\ b \in B\}.$$

Prove that every Borel subset of X is Carathéodory measurable with respect to μ^*.

> **Hint.** It is enough to prove that every closed set C is measurable. Fix a test set E. The case $\mu^*(E) = \infty$ is immediate, so assume it is finite. For $k \ge 0$, compare $E \cap C$ with the part of E lying outside the 2^{-k}-neighborhood of C. To control the omitted annuli, set
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> $$D_k \coloneqq \left\{x \in E \setminus C : 2^{-k-1} < \operatorname{dist}(x,C) \le 2^{-k}\right\}.$$
>
> Show separately that the even-indexed annuli are mutually separated, as are the odd-indexed annuli. Deduce that $\sum_k \mu^*(D_k) < \infty$, so the outer measure of the tail tends to zero.

4. (Uniqueness under σ-finiteness)

4.a (weight: 1.0) Let $\mathcal{P}$ be a π-system on X, and suppose that $\sigma(\mathcal{P}) = \mathcal{E}$. Let μ and ν be measures on $(X,\mathcal{E})$ such that

$$\mu(P) = \nu(P) \qquad \text{for every } P \in \mathcal{P}.$$

Assume that there are sets $P_n \in \mathcal{P}$ such that

$$X = \bigcup_{n=1}^{\infty} P_n \qquad \text{and} \qquad \mu(P_n) = \nu(P_n) < \infty \quad \text{for every } n.$$

Prove that $\mu = \nu$ on $\mathcal{E}$.